Your Thesis Title

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in Computer Science

Supervisor Dr. Mantis Toboggan

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(***Todo***: *Note that you can use the* `\todo` *command to add notes like these.*)

Here is an optional page environment you could use for things like:

- Your own custom preamble chapters (use `\chapterTitle` for titles!). If you want to add these to your table of contents, use the `\addChapterToTOC{text}` command and enable manual TOC entries in `preamble/thesis-details.tex` with `\enableManualTOCEntries`.
- *"This work is dedicated to..."*
- A copyright notice
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- Quotes
- List of publications
- An actual blank page

You can pass an optional parameter to this environment with the value `c` to centre this text vertically on the page.

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Signed: Benjamin Williams

Acknowledgements

Firstly, I want to thank somebody, and somebody else.[1] Here is another thing.

[1] Here is a footnote

Abstract

Here is the abstract for this thesis.

List of Tables

List of Figures

Table of Contents

Chapter 1

Introduction

1.1 Motivation

It is worth noting that this document was an **unofficial** thesis template for the University of Lincoln School of Computer Science. Since then, it has changed to encompass a broad range of theses. A decent template was needed for my thesis, so I made this. Then, other people liked it and decided to use it. Along the way, I've added in small features which people have suggested. If you have any suggestions or issues, please contact me – my email can be found at the top of `main.tex`! The following subsection in this document demonstrates some capabilities of the template (such as setting your thesis title, and so forth). I have not covered how to use LaTeX, as this assumes a working knowledge of the language.

If you are from another institution, then I want to thank you for your interest in my template! I've tried to make the template as lightweight as possible so you can modify it to your needs. You can easily swap out the logo (by changing logo.pdf or using \thesisLogoPath) and customise the title page contents using the commands listed on the next page. The template is also lightweight enough to enable you to change the typeface of the entire document. Again, if you have any questions please drop me an email as I love hearing your stories!

1.2 Task

The `csthesis` template has a number of definable variables which can be redefined to change the format of the thesis. These are listed below (on the next page).

Command	**Description**	**Example**
\thesisLogoPath	Sets the path to the logo for the title.	\thesisLogoPath{img/your-logo-here.jpg}
\thesisSubmissionDate	Manually sets the submission date of the thesis, for the title page.	\thesisSubmissionDate{July, 2019}
\thesisDegree	Sets the name of the degree for this thesis.	\thesisDegree{Doctor of Philosophy}
\thesisProgramme	Sets the name of your degree programme/subject.	\thesisProgramme{Biology}
\thesisSupervisor	Sets the name of your supervisor (optional).	\thesisSupervisor{Dr. Mantis Toboggan}
\thesisSecondSupervisor	Sets the name of your second supervisor (optional).	\thesisSecondSupervisor{Dr. Mantis Toboggan}
\thesisThirdSupervisor	Sets the name of your third supervisor (optional).	\thesisThirdSupervisor{Dr. Mantis Toboggan}
\thesisSchool	Sets the name of your school.	\thesisSchool{School of Chemistry}

\thesisCollege	Sets the name of your college.	\thesisCollege{College of Science}
\thesisUniversity	Sets the name of your university.	\thesisUniversity{University of Lincoln}

In addition, `\thesisSubmissionText{text}` overrides the submission statement text, so you can type whatever you'd like onto the front page. This is optional, though. For example, if you wish to erase it entirely, you can use `\thesisSubmissionText{}`. Example usage can be found in `preamble/thesis-details.tex`.

1.3 Placeholder

If you have included `[harvard]` in the document class command, then you will be able to cite things inline with parentheses. You can do this by using `\citep{citekey}`. It will print out something like this [**aad2012observation**]. Or alternatively, you can use `\cite{citekey}` to cite things like this [**chatrchyan2012observation**]. If you wish to use numeric style citing, just remove the `[harvard]` option from the `\documentclass` command.

This template uses BibLATEX for referencing, with a Biber backend. This is primarily due to the extensive features BibLATEX provides, along with the option of glossaries. If you want to customise the referencing style, you can either modify the template slightly to use different options, or use `\usepackage` again to reimport it. There's probably some commands to change its options after its been imported too.

If you wish to use numeric style citations instead, please remove the `[harvard]` option from your `\documentclass` command.

1.3.1 Ludography

This thesis template also contains an optional ludography. To use this, just put references into your bib file as usual with the game's details. Then, make sure `keywords` is set to `{game}`. This is what is used to determine which references are games, and which are actual papers. For a more elaborate example, see `bib/ludography.bib`.

Also, make sure that the `title` key is actually the author of the game, and the `author` is the title of the game. The reason this is swapped around is because BibLaTeX likes to print references out with the author first.

Then, just add `\printLudography` with an optional title argument to print out all citations like `\printLudography` or `\printLudography[Games]`.

You can also use the `ludography` environment if you wish to print out some text before the list of games is printed. An example of this can be seen in `main.tex`.

To cite games, you can `\cite` it like any other reference. However, if you want it to display the title instead of the standard referencing style, you can use `\citeGame` instead.

Here is an example of a cited game with a normal reference style: [**spaceinvaders**]. Ugh, pretty ugly. Instead, here the two are cited in the next sentence as games with `\citeGame`. Both **spaceinvaders** and **breakout** were games made by Atari. Much better!

Chapter 2

Fundamentals

2.1 Machine Learning

2.1.1 Face recognition tasks

Verification

Identification

2.1.2 Neural networks for vision

2.1.3 Embedding learning and similarity

2.2 Supervised learning

2.3 Performance metrics (verification)

2.4 Runtime metrics

2.5 Overview of models (architecture essentials)

2.6 Datasets and preprocessing

Chapter 3

Methods

3.1 Requirements)

3.2 Model and Dataset Selection

3.3 Concept and design

3.3.1 GUI Application

3.3.2 Validation and Benchmarking Pipeline

3.3.3 Performance Evaluation (CPU vs GPU)

3.3.4 Custom Dataset Integration

Here is a sentence, and you can see a nice picture in Figure 3.1. You can also use the `autoref` command to do this automatically, like this: Figure 3.1.

Also, a table can be found in Table 3.1. You should use a LaTeX table generator like `https://www.tablesgenerator.com/` if you want to make your life easier.

Table 3.1: Here is a table. The caption goes above like this.

First name	Last name	Age
Bob	Bobbington	24
Benth	Wavies	49
Joe	Bloggs	37
Billy	Bob	10

Figure 3.1: A picture of a duck.

Chapter 4

Results and Discussions

4.1 Experimental Setup

4.2 Quantitative Results (Accuracy, AUC, EER, TAR@FAR, etc.)

4.3 ROC Analysis (Image vs Video)

4.4 Comparison of Models

4.5 Discussion and Observations

Chapter 5

Conclusions

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